

Exercise 5 (i)

```
R=2;h=3;% radius and height in m
integral3(@integrand_conc, -R,R, -R,R,0,h)
```

```
ans = 7.7587
```

The total mass of the contaminant in the container is 7.7587 kg

Exercise 5 (ii)

```
f2 = @tank_vol;
T = 3600;
H = tank_height(T);
integral3(@integrand_conc, -2,2, -2,2,0,H)
```

```
ans = 17.2737
```

The mass in the tank is 17.2737kg

Exercise 5 part (iii)

```
function [c] = integrand_conc(x,y,z) % for part (i)
r = sqrt(x.^2+y.^2);
C0 = 0.1;
c = (C0*r.^2./sqrt(1+z.^2));
end

% for part (ii):

function [h] = tank_height(T) % calculate tank height
A=pi*4; % area of circle
h=integral(@qfunc,0,T)/A;
end

function [q] = qfunc(t)
Q0 = 0.1;w = pi/6;
q = Q0*(1+sin(w*t));
end

function [c] = timed_conc(x,y,z,t)
r = sqrt(x.^2+y.^2);
k = 0.001;
C0 = 0.1;
c = C0*exp(-k*t)*r.^2./(sqrt(z.^2+1));
end
```